

Haolin Ye

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EDUCATION BACKGROUND

The Pennsylvania State University

PhD in Electrical Engineering

Research area: Data Science and Artificial Intelligence; Signal and Image Processing

State College, PA

Aug 2024 - May 2029

Boston University

MS in Electrical and Computer Engineering

Boston, MA

Sept 2022 - Jan 2024

Core Courses: Deep Learning; Optimization Theory; Learning from Data; Enterprise Client- Server Software Systems Design

Shanghai Jiao Tong University, UM-SJTU Joint University

BE in Electrical and Computer Engineering, Minor in Data Science

Shanghai, China

Sept 2018 - Aug 2022

Core Courses: Probabilistic Methods; Programming and Elementary Data Structures; Introduction to Operating Systems; Introduction to Artificial Intelligence; Introduction to data Science; Computer Vision

Publications

Xu S, Huang Z, Zeng Y, Yan S, Ning X, Zhang Q, Ye H, Gu S, Shui C, Lin Z, Zhang H. HETHUB: A Distributed Training System with Heterogeneous Cluster for Large-Scale Models. arXiv preprint arXiv:2405.16256, 2024.

<https://arxiv.org/abs/2405.16256>

INTERNSHIP EXPERIENCE

Eriez Manufacturing Co.

Erie, PA

R & D Graduate Intern

May 2025- Aug 2025

- Developed orientation detection and orientation-robust deep learning models for improving electromagnetic metal detection accuracy.
- Designed a Conditional GAN to normalize signal variations caused by product orientation.
- Proposed and validated an Orientation-Aware ECF framework with improved generalization across diverse testing scenarios.

Shanghai Wuwen Xinqiong Intelligent Technology Co., Ltd

Shanghai, China

Intern at AI Research and Development Department, System Communication Engineer

May 2024- Aug 2024

- Built core components of the HETHUB distributed training system for large-scale models across heterogeneous clusters.
- Implemented various parallelism strategies including pipeline, tensor, data, and sequence parallelism.
- Developed a performance predictor and automatic parallel planner to optimize distributed training efficiency.
- Co-authored the paper “HETHUB: A Distributed Training System with Heterogeneous Cluster for Large-Scale Models.”

Institute for the Wireless Internet of Things, Northeastern University

Boston, MA

Research Assistant

May 2023- Sep 2023

- Based on the multi-task multi-sensor fusion framework BEVFusion, find approaches to reduce its FLOPS and training parameters to improve the model performance.
- Measured the FLOPS and parameter numbers of BEVFusion in the 3D detection task.
- Investigate the research tendency of quantization on Large Language Models and figure out the feasibility of further quantization of the activation layer.

SenseTime

Shanghai, China

Intern at the Mobile Intelligent Business Group, Software and System Development Engineer

Dec 2021 - May 2022

- Built the core technologies, platforms, products, and solutions of artificial intelligence and augmented reality on mobile terminals in the aspects of autonomous driving.
- Transformed and tuned the operating precision of the 3D LiDAR Object Detection network from Float32 to Float 16 to relieve the pressure of hardware memory.
- Tested the fine-tuned Object Detection network on different platforms and wrote corresponding guidelines.

Shanghai Taize Semiconductor Co., Ltd.

Shanghai, China

Intern at Chip Research and Development Department, Chip Research Intern

Aug 2021 - Dec 2021

- Led the intern team to complete the development of ResNet 18 and the optimization of Inception V3 on FPGA.
- Invoked and referred to a repository of previous built assembly functions to reimplemented ResNet18 and Inception V3 in C language.
- Tested the program on FPGA and assessed the precision of image recognition task.

Shanghai IC R&D Center

Shanghai, China

Intern at Artificial Intelligence Department, AI Research Intern

Jan 2021 - Apr 2021

- Comprehended deep convolutional neural networks within semantic image segmentation, used labeling tools to create data sets.
- Implemented the FCN and U-net and applied on edge detection task to evaluate the edge equality of produced chips automatically.
- Try other deeper neural networks such as Resnet 50 to improve the performance on segmentation.

RESEARCH EXPERIENCE

Information & Data Sciences Lab, Boston University

Boston, MA

Audio Transcription

Research Assistant, Advisor: [Brian Kulis](#)

Mar 2023- May

2023

- Investigated the application of the Google T5 model in audio transcription.
- Reimplemented and fine-tuned the T5 model using PyTorch to test its performance in audio transcription.

Intelligent Big Data System Lab, UM - SJTU JI

Shanghai, China

Deep Learning based on Blockchain Intelligent Network and Computing

Nov 2020 - Aug 2021

Research Assistant, Advisor: [Jie Li](#)

- Applied the core principles of Blockchain and combined the tech of Cloud Computing and Fog Computing into computing resources allocation.
- Applied Blockchain techniques in the related information fields, such as resource allocation or Internet of Vehicles.
- Leveraged Deep Reinforcement Learning (Q-learning) and Machine learning (Alternating Direction Method of Multipliers) to deal with computation offload and resource allocation.

Course PROJECT EXPERIENCE

Spam Detection Using Deep Learning and Boosting

Sept 2022 - Nov 2022

- Proposed a model that utilized word embedding and GloVe6B to convert a document into a trainable matrix which could extract features and adjust complexity dynamically, and achieved accuracy higher than 98%.

Implementation of a Simple Database

Sept 2021 - Dec 2021

- Realized a multi-threaded accelerated SQL database, and each table was divided into n sections to be computed in parallel, instructions were executed in parallel with read/write lock.

Analysis of Airbnb Data in Shanghai

May 2021 - Aug 2021

- Applied the Permutation Test to determine whether the price was influenced by the room type and reviews per month or not.

AWARDS & HONORS

2020, The 6th "Qian Xuesen Cup" Extracurricular Academic Science and Technology Works Competition for College Students in Shanghai Jiao Tong University - Second prize

2020, Midea Refrigerator School-Enterprise Joint Creative Competition Final - Top 5

2018, Shanghai Jiao Tong University RoboMaster Campus Competition - Champion

SKILLS

Programming Skills: C/C++, Python, R, Java, MATLAB, Verilog, MySQL, HTML, CSS

Frameworks and Libraries: Pytorch, Tensorflow, Pandas, Keras, Scipy, Sklearn

Developer Tools: Linux, Git, Docker, Shell, Latex, Markdown

Software: Clion, Pycharm, Vscode, Pscipe, Xilinx